Answer Key & Teacher Notes

Session 1 — Simplifying to Show Equivalence · Session 2 — Testing and Justifying Equivalence

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Simplifying to Show Equivalence

TALK PROMPTS

- What has to be true about two terms before you are allowed to combine them?
- In 9r + 7 + 6r + 8 = 15r + 15, both parts end in 15. Do the two 15s mean the same thing? Explain.

PARTNER TASK

Sorting Terms and Same or Not. One partner sorts the like terms in 4x + 12 + 3x + 5 and simplifies (7x + 17); the other checks by substituting x = 2 into both forms. Switch roles with 5n + 5n compared with 10n and with n^2.

WATCH FOR

- Combining a constant with a variable term (12x + 3 + 4x as 19x).
- Treating 5n + 5n as n^2 or $10n^2$ instead of 10n.
- Forgetting that a lone x has a coefficient of 1 (6 + 2k + k as 2k + 6).

SESSION 2 · TEACHER NOTES

Testing and Justifying Equivalence

TALK PROMPTS

- If two expressions match for one test value, does that prove they are equivalent? Why?
- You tested two values for the ticket formulas, 6t + 5 + 2t and 8t + 5. What else could you do to be sure they always agree?

PARTNER TASK

One Test Is Enough and Properties. One partner tests 3(k + 2) and 3k + 2 with k = 4 (18 vs. 14); the other names the property in 2 + (m + 11) = (2 + m) + 11 and tests m = 6 (19). Switch and justify 6t + 5 + 2t = 8t + 5 with properties.

WATCH FOR

- Deciding two expressions are equivalent after one matching test value.
- Distributing to only the first term (3(k + 2) as 3k + 2).
- Adding a constant to a coefficient (6 + 5j as 11j).

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.EE.A.3	1	2	1	1	2	2	3	2	2	1	1
S1 · Form B	6.EE.A.3	1	2	1	1	2	2	3	2	2	1	1
S2 · Form A	6.EE.A.4	1	2	1	1	2	2	2	3	2	1	1
S2 · Form B	6.EE.A.4	1	2	1	1	2	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

like terms: Terms with the same **variable** part.

combine: Add or subtract like terms into **one** term.

equivalent expressions: Expressions that have the same **value** for every number.

- Like terms have the same **variable** part; 4x and 4 are not like terms.
- Combine like terms by adding their **coefficients**: 4x + 9x = 13x.
- Two expressions are equivalent if they simplify to the **same** expression.

Example: The like terms with *b* are 9*b* and **3b**. Combine them: 9*b* + 3*b* = **12b**. Combine the constants: 4 + 1 = **5**. Answer: 9*b* + 4 + 3*b* + 1 = **12b + 5**

YOU TRY

1. a. 8a b. 5x + 3

2. Yes. 2n + 3n = 5n, so 2n + 7 + 3n = 5n + 7.

EXIT TICKET

1. a. 13x b. 5y + 5

2. 7k + 6

3. **A** (9*n* + 2)

PRACTICE FORM A

1. a. 8x + 8 b. 7m + 4 c. 7x + 17 d. 15r + 15 e. 10n f. 3k + 6

2. a. 3p and 2p b. 3p + 2p = 5p, so 5p + 10 c. Yes; Ana's expression simplifies to Luis's, so they are equivalent. d. 3p has the variable p and 10 does not, so they are not like terms.

3. 12m + 18

4. 11h + 15

5. Rashid is wrong. 4t and 16 are not like terms, so 4t + 16 is already simplified. Test t = 2: 4 × 2 + 16 = 24, but 20 × 2 = 40.

6. A: 5a + 10; B: 5a + 10; C: a + 15. A and B are equivalent.

7. a. 7p + 3 b. Yes, both are 7p + 3. c. Both equal 7 × 4 + 3 = 31. Equivalent expressions give the same value.

8. No. 4y and 5 are not like terms, so 4y + 5 cannot become 9y. For y = 2: 4 × 2 + 5 = 13, but 9 × 2 = 18.

9. Only 12x and 4x are like terms; the 3 has no x. 12x + 4x = 16x, so the answer is 16x + 3.

10. **B** (8*n* + 4)

11. **C** (5*m* and 5)

Self-check word **TERMS**

PRACTICE FORM B

1. a. 8y + 9 b. 7k + 5 c. 7x + 13 d. 12r + 15 e. 8n f. 4w + 9

2. a. 4t and 3t b. 4t + 3t = 7t, so 7t + 12 c. Yes; Maya's expression simplifies to Owen's, so they are equivalent. d. 4t has the variable t and 12 does not, so they are not like terms.

3. 14m + 16

4. 11h + 17

5. Lucia is wrong. 3t and 18 are not like terms, so 3t + 18 is already simplified. Test t = 2: 3 × 2 + 18 = 24, but 21 × 2 = 42.

6. A: 6b + 7; B: 6b + 7; C: b + 13. A and B are equivalent.

7. a. 9p + 5 b. Yes, both are 9p + 5. c. Both equal 9 × 8 + 5 = 77. Equivalent expressions give the same value.

8. No. 3y and 6 are not like terms, so 3y + 6 cannot become 9y. For y = 2: 3 × 2 + 6 = 12, but 9 × 2 = 18.

9. Only 10x and 6x are like terms; the 5 has no x. 10x + 6x = 16x, so the answer is 16x + 5.

10. **A** (8*a*)

11. **A** (like terms)

Self-check word **MATCH**

GUIDED NOTES

equivalent expressions: Expressions with the same value for **every** value of the variable.

counterexample: One test value that shows a claim is **wrong**.

Distributive Property: Multiply the number outside by **each** term inside the parentheses.

Commutative Property: You can change the **order** when you add or multiply.

- To test, substitute the **same** value into both expressions.
- One value that gives different results proves they are **not** equivalent.
- Matching values do not prove it; **justify** with properties.

Example: Try $x = 4$: $2 \times 4 + 8 = 16$ and $4 \times 4 =$ **16**. One match is not proof. Try $x = 1$: $2 \times 1 + 8 =$ **10** and $4 \times 1 = 4$. $10 \neq 4$, so $x = 1$ is a **counterexample**. Answer: They are **not** equivalent.

YOU TRY

1. a. 8 b. 7

2. m = 1: 5 and 5; m = 2: 10 and 6. Not equivalent (m = 2 is a counterexample).

EXIT TICKET

1. a. 12 b. 9

2. m = 1: 6 and 6; m = 4: 15 and 24. Not equivalent.

3. **C** (Distributive)

PRACTICE FORM A

1. a. 32 b. 32 c. 25. 2(y + 7) and 2y + 14 might be equivalent (the Distributive Property shows they are).

2. a. A: $6 \times 3 + 5 + 2 \times 3 = 29$; B: $8 \times 3 + 5 = 29$ b. A: 85; B: 85 c. Yes. Reorder and combine: 6t + 2t + 5 = 8t + 5. d. Commutative Property of Addition

3. 3(4 + 2) = 18; $3 \times 4 + 2 = 14$. $18 \neq 14$, so not equivalent.

4. a. Associative Property of Addition b. Both equal 19.

5. $6 + 5 \times 2 = 16$, but $11 \times 2 = 22$. $16 \neq 22$, so they are not equivalent; 6 and 5j are not like terms.

6. v = 1: 15 and 15; v = 7: 69 and 69. Yes; the Distributive Property gives 3(3v + 2) = 9v + 6.

7. $12 \times 5 - 4 \times 5 = 40$, but 8 = 8. Not equivalent. 12m − 4m = 8m; the m stays.

8. a. 18, 18, 10 b. 3(k + 4) and 3k + 12 c. Distributive Property: $3(k + 4) = 3 \times k + 3 \times 4 = 3k + 12$.

9. One matching value does not prove equivalence. For p = 2: $4 \times 2 + 6 = 14$, but $10 \times 2 = 20$, so they are not equivalent.

10. **B** ($2x + 10$)

11. **B** (not equivalent)

Self-check word **PROPERTY**

PRACTICE FORM B

1. a. 36 b. 36 c. 27. 4(y + 3) and 4y + 12 might be equivalent (the Distributive Property shows they are).

2. a. A: $4 \times 2 + 7 + 5 \times 2 = 25$; B: $9 \times 2 + 7 = 25$ b. A: 61; B: 61 c. Yes. Reorder and combine: 4b + 5b + 7 = 9b + 7. d. Commutative Property of Addition

3. $5(3 + 1) = 20$; $5 \times 3 + 1 = 16$. $20 \neq 16$, so not equivalent.

4. a. Associative Property of Addition b. Both equal 17.

5. $7 + 2 \times 3 = 13$, but $9 \times 3 = 27$. $13 \neq 27$, so they are not equivalent; 7 and 2j are not like terms.

6. v = 1: 12 and 12; v = 5: 44 and 44. Yes; the Distributive Property gives 4(2v + 1) = 8v + 4.

7. $15 \times 4 - 6 \times 4 = 36$, but 9 = 9. Not equivalent. 15n − 6n = 9n; the n stays.

8. a. 16, 16, 11 b. 2(k + 5) and 2k + 10 c. Distributive Property: $2(k + 5) = 2 \times k + 2 \times 5 = 2k + 10$.

9. One matching value does not prove equivalence. For p = 3: $4 \times 3 + 6 = 18$, but $10 \times 3 = 30$, so they are not equivalent.

10. **A** (not enough to prove equivalence)

11. **B** ($5a + 10$)

Self-check word **CONSTANT**

SHEET 1 · SIMPLIFYING TO SHOW EQUIVALENCE

1. a. 9a b. 6y c. 6x + 2 d. 3n + 11 e. 5b + 7 f. 7p + 9q g. 5m + 11 h. 11z

2. 2x + (x + 4) + (x + 1) = 4x + 5 cm

3. No. 3w + 3w = 6w, not $6w^2$ (for w = 2: 12 vs. 24).

4. 7x + 3 (combine 5x + 2x = 7x; the 3 has no x)

5. She combined the constant 4 with the x-terms (7 + 4 − 2 = 9). Only like terms combine: 7x − 2x = 5x, so the answer is 5x + 4.

SHEET 2 · TESTING AND JUSTIFYING EQUIVALENCE

1. a. 27 b. 27 c. 19. 3(x + 4) and 3x + 12 might be equivalent (the distributive property shows they are).

2. a. E b. N c. N d. E

3. Sample: n = 3 gives $n^2 = 9$ but 2n = 6, so they are not equivalent.

4. x = 4: 3x = 12 but x + 6 = 10, so they are not equivalent (one test value that gives different results is enough).

5. Distributive

6. **B** (7*p* + 21)

7. **D** (4(*y* + 8))

October 2026 conceptual additions — teacher key

Session 1 · Simplifying to Show Equivalence

Six x-tiles and eight unit tiles give 6x + 8. Like terms refer to the same variable part; unit constants cannot be added to the coefficient.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Six x-tiles and eight unit tiles give 6x + 8. Like terms refer to the same variable part; unit constants cannot be added to the coefficient.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Testing and Justifying Equivalence

First pair equivalent by distribution. Second not equivalent: x=1 gives 9 versus 6. One counterexample disproves equivalence; one matching test does not prove it.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: First pair equivalent by distribution. Second not equivalent: x=1 gives 9 versus 6. One counterexample disproves equivalence; one matching test does not prove it.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.